CLOUD HOSTING BUDGET ESTIMATE

PlutoniX Production Platform

A non-binding monthly operating estimate for an AWS Mumbai deployment.

REGION	CAPACITY	VOLUME
AWS Mumbai	25 concurrent builds	5,000 builds/month

EXPECTED MONTHLY BUDGET $3,400-$4,400 USD, including infrastructure and expected OpenAI model usage.

This estimate is designed for a multi-AZ launch serving approximately 100-300 active users. **It is not a binding quotation or invoice.**

Executive recommendation

Deploy PlutoniX as a managed control plane with isolated, queue-driven build workers. The existing local Docker Compose topology is suitable for development, but it should not be exposed directly as a public multi-user service.

- Serve the PlutoniX frontend and static previews through CloudFront and S3.
- Run the API as stateless ECS services across at least two availability zones.
- Queue builds in SQS and execute each build in an isolated, time-limited worker task.
- Store users, projects, builds, permissions, decisions, and snapshots in managed databases.
- Store generated files, media, exports, and workspace archives in S3.

Estimated monthly budget

Cost category	Expected range	What it covers
API and control plane	$90-$180	Multi-AZ ECS services for the PlutoniX API and orchestration control plane.
Build and preview workers	$150-$400	Approximately 5,000 isolated builds, queued overflow, and time-limited previews.
Aurora PostgreSQL	$180-$350	Project ownership, workflow state, build records, permissions, and snapshots.
Redis and queues	$60-$140	Live workflow state, event distribution, coordination, and SQS requests.
Network and load balancing	$100-$250	ALB, private connectivity, routing, and typical transfer.
Storage and delivery	$50-$180	S3, CloudFront, ECR, build artifacts, exports, and backups.
Security and operations	$100-$300	WAF, CloudWatch, audit logging, alarms, and backup operations.
Neo4j AuraDB	About $292	Business Critical minimum for the production agent relationship graph.
Infrastructure subtotal	$970-$1,940	Excludes model usage, engineering labour, taxes, and premium support.

AI model usage

Model usage is the largest variable. The estimates below use published GPT-5 pricing of $1.25 per million input tokens and $10 per million output tokens. Actual coding-agent usage depends on task complexity, retries, reviewers, context size, and caching.

Scenario	Tokens per build	Approx. monthly AI cost
Efficient	75k input + 15k output	$1,220
Expected	150k input + 30k output	$2,438
Heavy	300k input + 60k output	$4,875

COST-CONTROL OPPORTUNITY Use smaller models for classification, routine edits, and low-risk review. Reserve GPT-5-class execution for complex or high-risk work. A hybrid route could reduce expected AI usage to roughly $700-$1,500 per month.

Required production work

1. Remove Docker-socket access and replace host-created preview containers with isolated cloud tasks.
2. Replace local JSON, JSONL, logs, workspaces, and project registries with managed persistence.
3. Cryptographically verify Google identity tokens and enforce ownership on every project resource.
4. Convert synchronous generation into durable queued jobs with status, cancellation, retries, and idempotency.
5. Build production frontend assets; do not operate public Vite development servers.
6. Add quotas, rate limits, audit logs, malware scanning, worker network controls, monitoring, and tested backups.

Architecture at a glance

Layer	Recommended service	Responsibility
Edge	Route 53, ACM, CloudFront, WAF	DNS, TLS, caching, and perimeter protection.
Application	ALB + ECS Fargate	Authenticated API and orchestration control plane.
Build execution	SQS + isolated ECS tasks	Sandboxed agent execution with bounded resources.
Data	Aurora PostgreSQL + Redis	Durable metadata and short-lived coordination state.
Knowledge	Neo4j AuraDB + vector store	Agent relationships and reusable memory.
Artifacts	S3 + CloudFront	Generated files, exports, media, snapshots, and previews.

Assumptions and exclusions

- AWS Mumbai, multi-AZ operation, 100-300 active users, and 25 concurrent builds.
- Approximately 5,000 build instructions per month and a ten-minute average worker duration.
- Static previews are preferred; dynamic previews expire after 30 minutes of inactivity.
- Amounts exclude implementation labour, GST or other taxes, premium support, unusually high egress, and third-party observability subscriptions.
- Provider rates and exchange rates may change; confirm the final design in the AWS Pricing Calculator before purchase.

VALIDITY Budgetary estimate valid for planning purposes for 30 days from 5 July 2026. Final costs are usage-based and charged directly by service providers.

Reference pricing

Amazon Lightsail instance bundles:
https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-bundles.html

AWS Fargate pricing: https://aws.amazon.com/fargate/pricing/

AWS Elastic Load Balancing pricing: https://aws.amazon.com/elasticloadbalancing/pricing/

OpenAI GPT-5 developer pricing: https://openai.com/index/introducing-gpt-5-for-developers/

Neo4j AuraDB pricing: https://neo4j.com/pricing/

Budget decision

For a private beta, a single hardened virtual machine can reduce initial infrastructure cost significantly. For the selected 25-concurrent-build production target, the managed architecture above is the safer baseline because it separates users, build workers, data, and credentials.